Stable Mode-Hop-Free Feedback-and-Feedforward Loop to Extend the Tuning Range of External-Cavity Diode Lasers

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Introduction

Tuning External-Cavity Diode Lasers (ECDLs) often requires a simultaneous adjustment of the cavity length and the grating angle to ensure the optical length stays matched in order to avoid mode hopping. Traditionally [1][2], this is achieved locally by adjusting the piezoelectric transducer (PZT) voltage and the injection current with feedback loops using error signals. However, this method is not applicable for cases such as high-speed scanning where the thermal relaxation time becomes significant, or wide-range tuning where PZT hysteresis accumulates. We propose a quasi-global linear parameter-varying (LPV) control model, which can be integrated to the traditional feedback method, to study and generalize the dynamics of ECDLs. By considering temperature, PZT hysteresis, and the geometric configuration of ECDLs, we derive the phase constraint equation and the parameter-varying state-space model that predicts and compensates for error. The model achieves a broader mode-hop-free (MHF) tuning condition by incorporating feedback and feedforward control, and a numerical simulation comparing the standard proportional feedback loop to the proposed feedback-and-feedforward loop is presented to demonstrate the stability of the LPV method and its efficiency in reducing transient phase errors.

Theoretical Prediction

The total phase of an ECDL is governed by

$$\Phi_i(\nu, I, T) + \Phi_e(\nu, L\,[\theta(V, \phi, y)]) + \Phi_r(\nu)$$

where Φ_i is the internal phase, Φ_e is the external phase, Φ_r is the phase of reflection; ν is the laser frequency, I is the injection current, T is the temperature, and $L\,[\theta(V(y), \phi)]$ is the functional of the optical path between the grating and the facet that depend on the angle of incidence, a function of PZT voltage and coarse adjustment. The PZT displacement has a significant, highly non-linear hysteresis, thus one must consider a function of hysteresis y to account for such behavior. The total phase must be an integer multiple of 2π to coherent resonance. Therefore, we define the constraint

$$F(\nu, I, T, L) = \Phi_i(\nu, I, T) + \Phi_e(\nu, L\,[\theta(V, y)]) + \Phi_r(\nu) - 2\pi N$$

to analyze the system. Differentiating the constraint by time and manipulating the expression yields

$$\dot{\nu} = \mathcal{D}^\nu_I\, \dot{I} + \mathcal{D}^\nu_T\, \dot{T} + \mathcal{D}^\nu_L\, \mathcal{D}^L_V\, \dot{V},$$

where

$$\mathcal{D}^\nu_I \equiv -\frac{\partial_I F}{\partial_\nu F},\ \mathcal{D}^\nu_L \equiv -\frac{\partial_L F}{\partial_\nu F},\ \mathcal{D}^\nu_T \equiv -\frac{\partial_T F}{\partial_\nu F},\ \mathcal{D}^L_V \equiv \partial_\theta L\, \partial_y \theta\, \partial_V y.$$

Consider the following state-space equation [3]

$$\dot{\mathbf{x}}(t) = \mathrm{A}(t)\mathbf{x}(t) + \mathrm{B}(t)\mathbf{u}(t) + \mathrm{C}(t)\dot{\mathbf{u}}(t),$$

where the matrices $\mathrm{A}(t)$ and $\mathrm{B}(t)$ represent drift terms. The state equation can be written as an affine form of $\dot{\mathbf{u}}$,

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}, \mathbf{u}) + g_1(t)\dot{I} + g_2(t)\dot{V},$$

where

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \mathcal{D}^\nu_I\, \dot{I} + \mathcal{D}^\nu_L\, \mathcal{D}^L_V\, \dot{V} + \mathcal{D}^\nu_T\, (A_T T + B_T I) \\ \mathcal{D}^L_V\, \dot{V} + A_L L \\ A_T T + B_T I \\ \partial_V y\, \dot{V} + A_y y \end{bmatrix}.$$

As a theoretical justification, we will show that the LPV control method proposed in this study is necessary to achieve a globally stable wide MHF tuning range. Identifying the state vector $\mathbf{x} = [\nu, L, T, y]^T$, the control vector $\mathbf{u} = [I, V]^T$, and the coefficient matrices $g_1(t)$ and $g_2(t)$, we define a vector field operation $X_i = \partial_{\dot{u}_i} \dot{\mathbf{x}}$, the tangent vectors on the control manifold; where $\partial_{\dot{u}_i}$ denotes differentiation with respect to the i-th component of $\dot{\mathbf{u}}$. That is,

$$X_1 = \begin{bmatrix} \mathcal{D}^\nu_I \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad X_2 = \begin{bmatrix} \mathcal{D}^\nu_L\, \mathcal{D}^L_V \\ \mathcal{D}^L_V \\ 0 \\ \partial_V y \end{bmatrix}.$$

The Lie bracket is given by

$$(\mathcal{L}_{X_1} X_2)_j = [X_1, X_2]_j = \left(\frac{\partial X_2}{\partial \mathbf{x}}\, X_1 - \frac{\partial X_1}{\partial \mathbf{x}}\, X_2 \right)_j,$$

where a non-zero commutation of the Lie derivatives imply the non-integrability of the system. In other words, this indicates that the control manifold has a fundamental curvature which makes the system path-dependent. One may obtain

$$(\mathcal{L}_{X_1} X_2)_1 = \partial_\nu(\mathcal{D}^\nu_L\, \mathcal{D}^L_V) \cdot (\mathcal{D}^\nu_I) - \partial_\nu(\mathcal{D}^\nu_I) \cdot (\mathcal{D}^\nu_L\, \mathcal{D}^L_V),$$

$$(\mathcal{L}_{X_1} X_2)_2 = \partial_\nu(\mathcal{D}^L_V) \cdot (\mathcal{D}^\nu_I), \qquad (\mathcal{L}_{X_1} X_2)_4 = \partial_\nu(\partial_V y) \cdot (\mathcal{D}^\nu_I),$$

which are necessarily non-zero. The LPV control method proposed effectively linearizes the curved manifold, unlike a fixed-gain PID controller, ensuring enhanced stability.

To construct the feedforward control model, we recall Bragg's law and obtain

$$\dot{\nu}_g = -\frac{c}{2d}\, \frac{\cot^2\theta \cos\theta}{l}\, \dot{L} \equiv -M(\theta)\dot{L},$$

where d is the grating spacing and l is the distance between the pivot and the facet. We define the error frequency slope

$$\dot{\nu}_e = \dot{\nu} - \dot{\nu}_g,$$

and for a particular application where $\dot{\nu}_e = 0$, the constraint gives the necessary time derivative of the feedforward current,

$$\dot{I}_{ff}(t) = -\frac{1}{\mathcal{D}^\nu_I}\, [\mathcal{D}^L_V(\mathcal{D}^\nu_L + M(\theta))\dot{V} + \mathcal{D}^\nu_T B_T I + \mathcal{D}^\nu_T A_T T + M(\theta) A_L L].$$

The full feedback-and-feedforward model is given by

$$I(t) = I_0(t) + K_p\, e(t) + K_i \int_0^\tau e(t)\,\mathrm{d}t + \int_0^\tau \dot{I}_{ff}(t)\,\mathrm{d}t,$$

or equivalently,

$$\Delta I = \Delta I_0 + K_p(e_n - e_{n-1}) + K_i\, e_n\, \Delta t + I_{ff}\, \Delta t,$$

where $I_0(t)$ acts as a base current. Due to the non-zero Lie bracket, $I_0(t)$ need not be a constant.

Numerical Analysis

Consider

$$\dot{I}_{ff}(t) \approx -\frac{\mathcal{D}^L_V(\mathcal{D}^\nu_L + M(\theta))}{\mathcal{D}^\nu_I}\, \dot{V},$$

in which both relaxation terms and the Joule heating term are neglected. The differential current is given by

$$\Delta I_{ff} = -G^{-1}(\nu)\, H(V) \left(-\frac{\nu}{L} + \frac{c}{2d}\, \frac{\cot^2\theta \cos\theta}{l} \right) \Delta V,$$

where $G^{-1}(\nu)$ is the inverse slope frequency-current slope, equivalently, $(\partial\nu/\partial I)^{-1}$; and $H(V)$ is equivalent to $l \sec^2\theta\, (\partial\theta/\partial V)$.

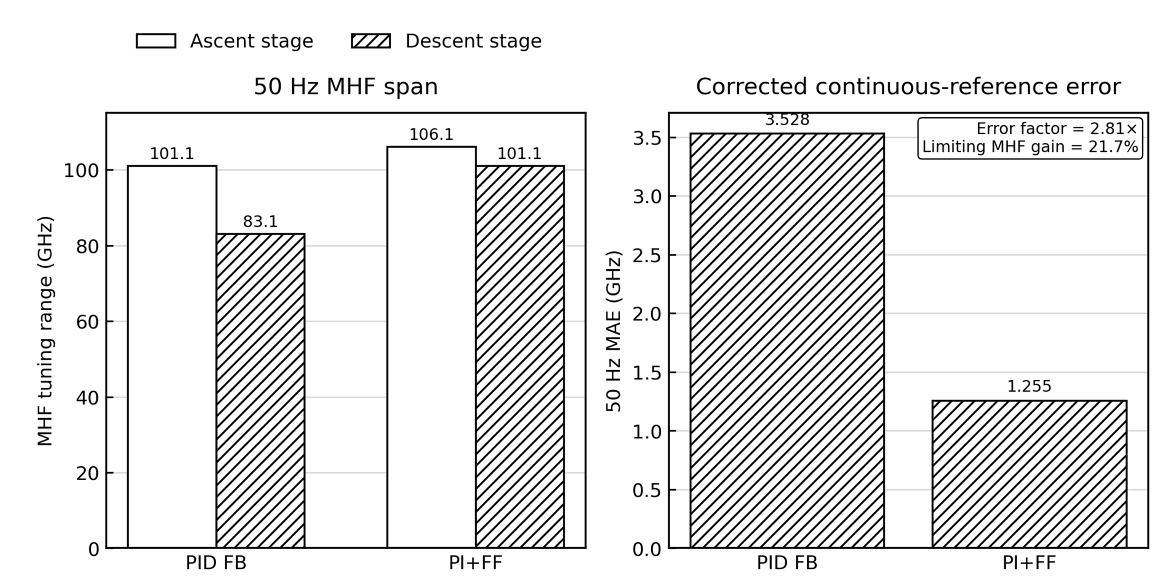

Figure 1: MHF tuning range comparisons between PID control and PI with FF (PIFF) control at 50 Hz scanning frequency. Left panel denotes ascent and descent MHF tuning range with PIFF compared to PID, and right panel denotes error for descent.

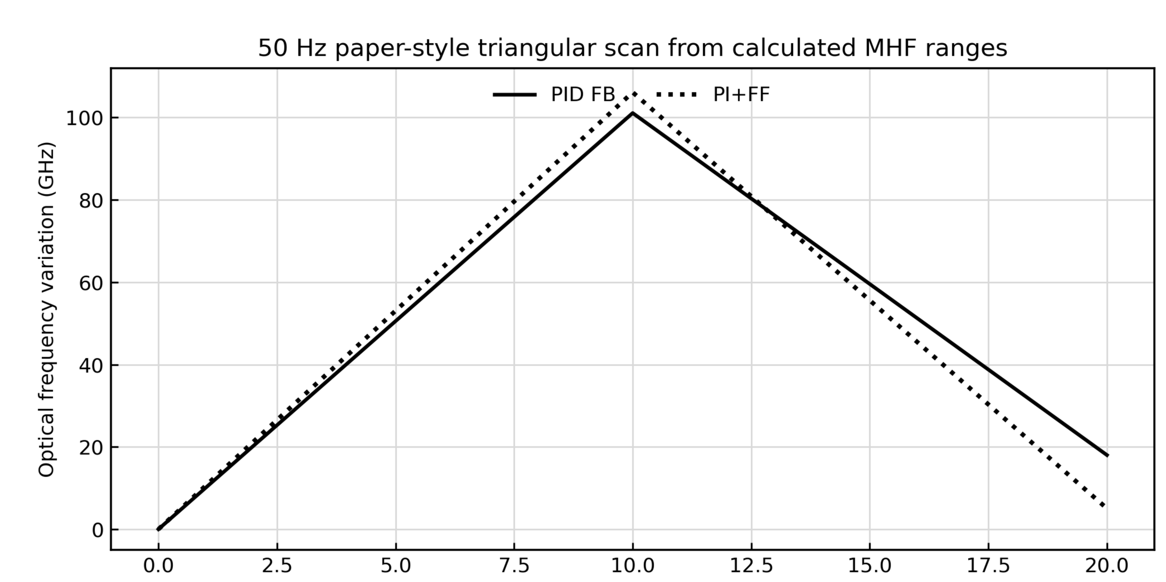

Figure 2: Continuous optical-frequency-variation comparison for PID vs. PIFF at 50 Hz scanning frequency. The rising half-cycle height is ascent-stage MHF tuning range, while the downward change from the peak gives the descent-stage MHF tuning range.

We use an approximate physically structured reduced-order simulation of the uncoated ECDL scan in Zhu. et al [4], retaining the core synchronous-tuning idea that the internal-cavity selection frequency and the external-cavity selection frequency must remain matched closely enough that their difference stays within the external-cavity free spectral range criterion for mode-hop-free operation. The external cavity is driven by a triangular scan and includes branch-dependent PZT hysteresis, a dominant flexure response, and a higher-frequency vibration mode. The internal cavity is driven through the injection-current path and includes a small thermal-memory term so that residual lag can accumulate during faster scans. For the baseline case, the controller is a PID feedback loop acting on the external-cavity motion, consistent with the paper's displacement-feedback strategy. Our proposed controller keeps that external-cavity stabilization idea but adds a feedforward current correction through the internal-cavity path, implemented here as a PIFF branch derived from the phase constraint and inverse frequency-current slope developed in the earlier sections. We use this simulation to calculate the MHF tuning range numerically and to compare tracking error across controllers. In this reduced simulation, the feedforward current branch produces a non-negligible improvement over PID feedback alone, specifically at 50 Hz scanning frequency where hysteresis, vibration, and thermal lag are more important.

Conclusion

In our simulation, the proposed PIFF controller improves both the achievable mode-hop-free tuning range and the frequency-tracking accuracy relative to the PID feedback baseline in previous work. At 50 Hz, the fixed PID baseline gives an MHF tuning range of approximately 101.1 GHz in the ascent stage and 83.1 GHz in the descent stage, with a corrected continuous-reference MAE of about 3.53 GHz. These metrics are within 3% of the actual values in the paper, emphasizing the robustness of the simulation. Under the same model, the optimized PIFF controller increases the MHF tuning range to about 106.1 GHz in ascent and 101.1 GHz in descent, while reducing the MAE to about 1.25 GHz. This corresponds to an improvement of roughly 21.7% in the limiting MHF span and an error-reduction factor of about 2.8. The improvement arises because feedforward current correction acts directly on the internal-cavity path, reducing internal/external cavity mismatch in regions where hysteresis, vibration, and thermal lag limit a feedback-only controller, proving the efficacy of our model over PID feedback methods.

References

[1] Repasky et al, Appl. Opt. 45, 9013-9020 (2006)
[2] Gong et al, Appl. Opt. 53, 7878-7884 (2014)
[3] The state-space equation can be expressed as a standard state equation, $\dot{\mathbf{v}} = \mathbf{A}\mathbf{v} + (\mathrm{AC} + \mathrm{B})\mathbf{u}$, where $\mathbf{v} = \mathbf{x} - \mathrm{C}\mathbf{u}$.
[4] Zhu et al, Appl. Phys. B 130, 200 (2024)